
How Much of BERT Does Text Classification Need? A Controlled Comparison with DistilBERT

Jose Huamani, Renzo Felix, Diego Quispe

Universidad de Ingeniería y Tecnología (UTEC), Lima, Peru
{jose.huamani, renzo.felix, diego.quispe.ap}@utec.edu.pe

Abstract

DistilBERT promises most of BERT’s accuracy at about half the inference cost, and we test that claim on SST-2, AG News and Yelp Polarity with one recipe shared by both models. With 61% of BERT’s parameters, DistilBERT keeps 98.9% to 99.5% of its test accuracy. The gap depends on the task. It reaches 1.04 points on SST-2 and disappears on AG News. On a controlled grid of nine batch and length settings, DistilBERT is 1.86 to 2.15 times faster and needs 23% to 38% less GPU memory. An ablation of six classifier configurations on DistilBERT finds that head width and depth barely matter, since five heads with a trainable encoder land within 0.13 points of mean validation F1, while freezing the encoder costs 3.4 to 9.1 points. Moved to BERT, the winning linear head widens the SST-2 gap to 1.95 points and leaves AG News at 0.11.

1 Introduction

BERT [Devlin et al., 2019] made pre-training followed by fine-tuning the default recipe for text classification. Its cost shows up at inference, where a deployed model spends most of its life and every request crosses 12 transformer layers. DistilBERT [Sanh et al., 2019] halves the depth through knowledge distillation and reports 97% of BERT’s GLUE score.

We set one rule for this project: the model is the only thing allowed to differ between two columns of any table, so both networks share the training loop, splits and seed. The rule broke once, for latency. The per-run timing logged after each fine-tuning claimed that BERT answered faster than DistilBERT on SST-2 (3.16 against 3.50 ms), which a network with twice the layers cannot do. We threw those numbers away and measured both models again on a fixed grid (Section 3.2).

We fine-tune both models on SST-2 [Socher et al., 2013, Wang et al., 2019], AG News and Yelp Polarity [Zhang et al., 2015] and report accuracy, macro precision, recall and F1, along with parameters, latency and GPU memory. An ablation on DistilBERT varies the classifier head and freezes the encoder, and the best configuration is re-run on BERT with every metric.

2 Approach

Models. We start from the public bert-base-uncased (12 layers, 109.5 M parameters) and distilbert-base-uncased (6 layers, 67.0 M) checkpoints on the HuggingFace Hub. DistilBERT keeps every second layer, the hidden size of 768 and the vocabulary, and drops the token-type embeddings and the pooler. Its parameter count falls by 39% instead of 50% because the 23.4 M-weight embedding matrix survives intact.

Data. A single adapter maps the three datasets to one schema (text, label) and three splits. GLUE ships the SST-2 test split without labels, so the official validation set (872 sentences) acts as our test set and 10% of the training data becomes validation. AG News (4 classes, 7,600 test

Table 1: Base runs with each library’s default head. Test metrics in %. Precision, recall and F1 are macro averages. Latency and memory come from the controlled benchmark at batch 1 and the dataset’s input length. Bold marks the better model per dataset.

Dataset	Model	Test (%)				Efficiency			
		Acc	Prec	Rec	F1	Params (M)	Latency (ms)	Memory (MB)	Train (min)
SST-2	BERT	92.32	92.32	92.31	92.31	109.5	5.56	430	4.2
SST-2	DistilBERT	91.28	91.34	91.25	91.27	67.0	2.60	269	2.7
AG News	BERT	94.41	94.43	94.41	94.41	109.5	5.61	433	9.1
AG News	DistilBERT	94.50	94.52	94.50	94.50	67.0	2.61	272	5.5
Yelp	BERT	96.52	96.52	96.52	96.52	109.5	5.11	436	16.0
Yelp	DistilBERT	96.06	96.06	96.06	96.06	67.0	2.74	275	8.2

items) and Yelp Polarity (2 classes, 38,000) have no validation split either, and we hold out 10% of training with a fixed seed. Yelp is subsampled to 100,000 training and 20,000 validation reviews out of 560,000. The subset is identical for both models. Inputs are padded to 64, 128 and 256 tokens for SST-2, AG News and Yelp.

Training recipe. AdamW with learning rate 2×10^{-5} and weight decay 0.01 (none on biases or LayerNorm), 10% linear warm-up and linear decay, 2 epochs, batch 32, fp16 mixed precision, gradients unscaled before clipping at norm 1.0, seed 42. We keep the checkpoint with the best validation macro-F1 and touch the test set once. The base runs use each library’s default head. For BERT that is the pooler (tanh) plus a linear layer, and for DistilBERT a `pre_classifier` with ReLU.

Ablation head. The two default heads differ, so the ablation replaces both with one module on the [CLS] state: a stack of Linear–ReLU–Dropout(0.1) layers followed by a linear output over C classes. We vary the width (128, 768 or 2048 units), the depth (zero, one or two hidden layers) and freezing. A frozen encoder is also removed from the optimizer, so weight decay cannot touch it. Its head trains at 10^{-3} , since a head learned from scratch barely moves at 2×10^{-5} . The configuration with the highest validation F1 averaged over the three datasets wins.

Efficiency measurement. Latency is the mean of 50 forward passes after 20 warm-up passes, with `torch.cuda.synchronize()` around each one, in fp32 on an NVIDIA RTX A6000. Both models run in the same process over a grid of batch sizes $\{1, 8, 32\}$ and lengths $\{64, 128, 256\}$, which also records peak GPU memory. Tables quote the point at batch 1 and the dataset’s input length.

3 Experiments

The study has 24 runs: six base runs, 18 ablation runs on DistilBERT and three runs of the winning configuration on BERT, all on one RTX A6000.

3.1 Performance

Table 1 holds the six base runs. BERT wins on SST-2 by 1.04 accuracy points and on Yelp by 0.46. On AG News the sign flips to DistilBERT by 0.09 points, which is seven test items out of 7,600. We read that as a tie. DistilBERT’s 91.28% on SST-2 matches the 91.3 reported by Sanh et al. [2019], a sign that the splits leak nothing and the recipe is sound.

In Figure 1 both models trace almost the same loss path. On SST-2, training loss keeps falling in the second epoch while validation loss stalls near 0.17, which is why we select checkpoints on validation F1.

3.2 Efficiency

Figure 2 puts size, accuracy and latency in one picture. The DistilBERT column sits 42.5 M parameters to the left of BERT with bubbles half as large, and the vertical distance between each pair is the accuracy price. On the controlled grid (Figure 4) DistilBERT is faster at all nine points, by $1.86 \times$

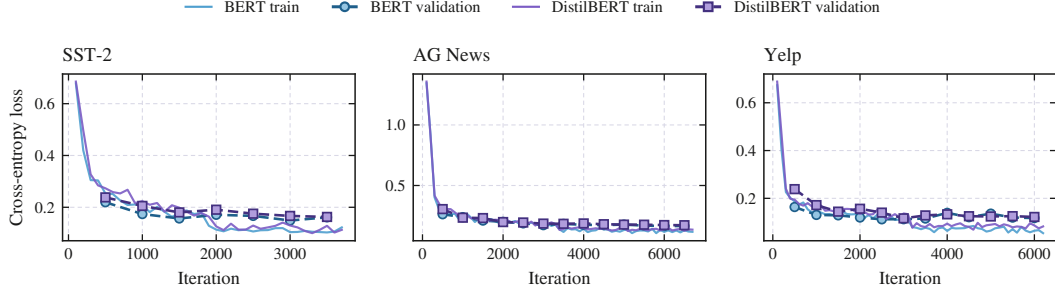


Figure 1: Training loss (logged every 100 steps) and validation loss (every 500 steps) against optimizer iterations for the six base runs.

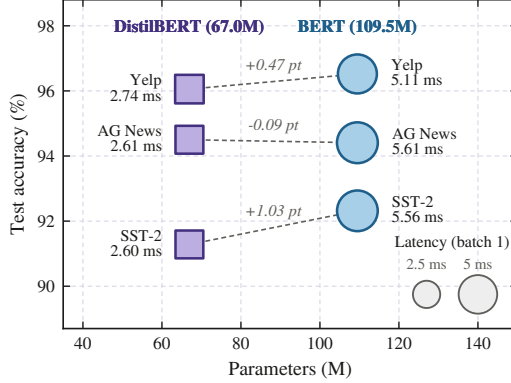


Figure 2: Parameters against test accuracy. Bubble area is latency at batch 1 and the dataset’s input length.

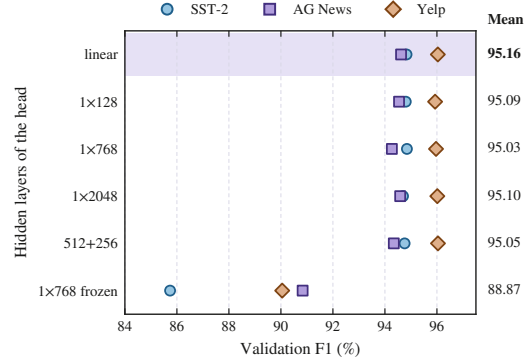


Figure 3: Validation F1 of the six DistilBERT configurations. The lilac row is the selected one. The right column is the mean over datasets.

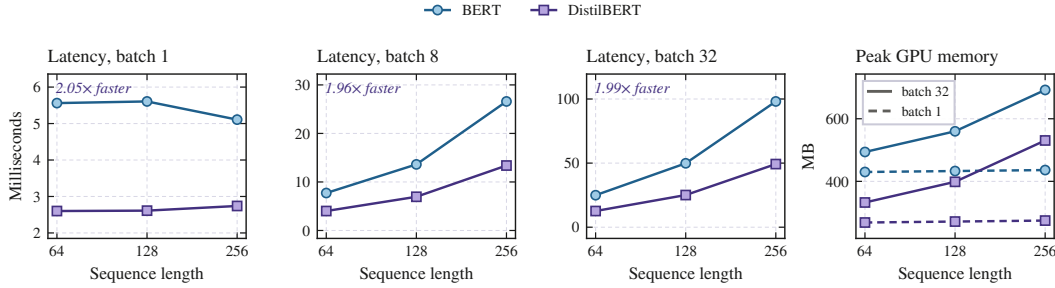


Figure 4: Controlled benchmark (RTX A6000, fp32, one process). Three latency panels, one per batch size, with the mean speed-up of DistilBERT written in each. The last panel shows peak GPU memory at batch 1 and batch 32.

to 2.15 $\times$. At batch 32 and 64 tokens it processes 2,535 samples per second against 1,282, and its peak memory is 62% of BERT’s at batch 1 and 77% at batch 32 with 256 tokens. Training the three datasets took 16.4 minutes against 29.3.

3.3 Ablation study

Table 2 and Figure 3 cover the six configurations. The five heads with a trainable encoder span 95.03 to 95.16 mean validation F1, and their order is unstable across datasets (h768 is the best on SST-2 and the worst on average). The frozen encoder drops to 88.87. Against h768, which has the same head, it loses 9.1 points on SST-2, 5.9 on Yelp and 3.4 on AG News, in exchange for training 0.6 M parameters in 6.0 minutes. The linear head wins on the mean, and it is also the smallest and fastest to train.

Table 2: Ablation on DistilBERT. Validation F1 in %. The lilac row is the selected configuration (highest mean), and bold marks the best value per column. Time is the sum over the three datasets.

Validation F1 (%)								
Config	Head	Encoder	SST-2	AG News	Yelp	Mean	Trainable (M)	Train (min)
lineal	768 → C	trained	94.83	94.62	96.03	95.16	66.4	15.3
h128	768 → 128 → C	trained	94.80	94.54	95.93	95.09	66.5	15.9
h768	768 → 768 → C	trained	94.85	94.27	95.96	95.03	67.0	16.3
h2048	768 → 2048 → C	trained	94.70	94.59	96.01	95.10	67.9	16.3
c2	768 → 512 → 256 → C	trained	94.76	94.34	96.04	95.05	66.9	15.9
frz	768 → 768 → C	frozen	85.74	90.84	90.05	88.87	0.6	6.0

Table 3: Linear head (lineal) on both models, same recipe. Test metrics and validation F1 in %, efficiency as in Table 1. Bold marks the better model per dataset.

Performance (%)							Efficiency			
Dataset	Model	Acc	Prec	Rec	F1	Val F1	Params (M)	Latency (ms)	Memory (MB)	Train (min)
SST-2	BERT	92.89	92.94	92.86	92.88	95.41	109.5	5.56	430	4.2
SST-2	DistilBERT	90.94	90.99	90.91	90.93	94.83	66.4	2.60	269	1.7
AG News	BERT	94.53	94.56	94.53	94.53	94.81	109.5	5.61	433	9.0
AG News	DistilBERT	94.42	94.43	94.42	94.42	94.62	66.4	2.61	272	5.4
Yelp	BERT	96.53	96.53	96.53	96.53	96.61	109.5	5.11	436	16.0
Yelp	DistilBERT	96.10	96.10	96.10	96.10	96.03	66.4	2.74	275	8.2

3.4 Best configuration on BERT

With the linear head (Table 3) BERT wins all three datasets, by 1.95 points on SST-2, 0.43 on Yelp and 0.11 on AG News.

4 Analysis

Task dependence. Topic labels in AG News hinge on vocabulary that six layers already capture, and the per-class errors show it. BERT sends 140 Business stories to Sci/Tech and 90 the other way. DistilBERT makes 140 and 92. The extra depth finds nothing more in those items. Short SST-2 sentences full of negation and contrast are where the extra layers pay off.

Head versus encoder. Five heads, from one linear layer to two hidden layers, fit inside 0.13 points. Fine-tuning shapes the [CLS] vector until a linear boundary separates the classes, which leaves extra head layers nothing to do. The linear head even lifts BERT above its own base run on SST-2 (92.89 against 92.32 accuracy), so the default tanh pooler adds nothing here.

Latency regimes. At batch 1 latency stays flat from 64 to 256 tokens (BERT 5.56, 5.61 and 5.11 ms) because the GPU is waiting on kernel launches. From batch 8 on, latency tracks the input length. DistilBERT wins both regimes, with half the kernels to launch in the first and half the arithmetic in the second. Its memory edge shrinks as activations outgrow the weights.

Limitations. All runs use seed 42, so gaps of a few tenths are ties and we cannot give intervals. Two epochs sit at the low end of what Devlin et al. [2019] recommend. Yelp is subsampled for both models alike. Five base runs predate the logging of validation F1, hence its absence from Table 1.

5 Conclusion

DistilBERT is the sensible default for these tasks: 98.9% to 99.5% of BERT’s accuracy with 61% of the parameters, two thirds of the memory and half the latency. BERT’s margin is what a team has to price. On AG News, 0.11 points will rarely justify doubling the inference bill. On SST-2, 1.95 points might. For either model the classifier can stay a single linear layer on [CLS], which matched the 2048-unit and two-layer heads within 0.13 points.

References

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *NeurIPS EMC² Workshop*, 2019.
- Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of EMNLP*, pages 1631–1642, 2013.
- Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of ICLR*, 2019.
- Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.